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EARLY 'UMLAUT' PHENOMENA IN THE GERMANIC LANGUAGES

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Not only the evidence of Gothic, but also that of North and West Germanic, speaks against the postulation of Proto-Germanic 'umlaut' allophones (§1). Furthermore, since even the results of the 'earlier umlaut' phenomena affecting **i e u/* in the various dialects point to the rise of allophonic variations in partly different environments, all the pre-literary vowel modifications produced by both vocalic and consonantal influences should be ascribed to the prehistory of the individual Germanic languages (§§2–3). The scarcity of forms with */e/* by *a*-umlaut of **i/* is explained through the assumption that the assimilation exerted by **[a]* on **i/* was widely resisted in order to avoid a merger of **i/* with **e/* (§2).

1. ALLEGED PROTO-GERMANIC 'UMLAUT' ALLOPHONES. Although no vowel modifications comparable to those attested in the other Germanic languages can be shown to have occurred in Pre-Gothic,¹ the suggestion that at least certain early 'umlauts'² should be ascribed to some stage of Proto-Germanic is by no means new.³ These 'earlier (or Proto-Germanic) umlauts'—as distinct from 'later (or idiolingual) umlauts'⁴—include:

(1) the change **e/* > **i/* before a high front vowel or semivowel (*i*-umlaut), as well as before a cluster of nasal plus consonant:⁵ PIE **/med^hjos/* (cf. Lat. *medius*) > PGmc. **/meðjaz/* > OIc. *mīþr*, OE *midd*, OFris. *midde*, OS *middi*, OHG *mitti* (Go. *midjis*) 'mid'; PIE **/wentos/* (cf. Lat. *ventus*) > PGmc. **/wenðaz/* > OIc. *vindr*, OE OFris. OS *wind*, OHG *wint* (Go. *winds*) 'wind';

(2) the changes **i/* > **e/*, **u/* > **[o]* (> **o/*)⁶ before a low or mid vowel,

¹ See Cercignani 1979a. But the forms adduced by Wienold 1967 against umlaut in Gothic must be rejected as inconclusive (see Antonsen 1969: 68–70, 1972: 140). As for a special *a*-umlaut across internal open juncture, proposed by Bennett 1967 to explain the */e/* of the Gothic reduplicating syllable, it is counter-indicated by the fact that the same assimilatory influence invariably failed to operate across internal close juncture (see Cercignani 1979b).

² When enclosed in quotation marks, the term 'umlaut' is here used to cover not only vowel assimilation proper, but also those cases of consonantal influence which are sometimes associated—for the sake of terminological convenience or otherwise—with cases of genuine umlaut (cf., e.g., fnn. 5 and 11, below).

³ See, e.g., Streitberg (1896, §94). For a rejection of Streitberg's view (§71) that spellings like *Gotones/Gothones* in Tacitus point to the occurrence of umlaut phenomena in Pre-Gothic, see Bennett (1952: 340–41).

⁴ See, e.g., Sonderegger (1959: 3–5). It would seem that a suitable translation of Ger. *einzel-sprachlich* is 'idiolingual(ly)', which I here propose as a new coinage. Ger. *einzel-dialektisch* can similarly be rendered by 'idiodialectal(ly)', as distinct from 'idiolectal(ly)' (Ger. *idiolektisch*).

⁵ This latter case is obviously caused by consonantal influence; but since the result is the same as for the *i*-umlaut of **e/* (and both are traditionally ascribed to Proto-Germanic), the term '(earlier) *i*-umlaut' is sometimes used to cover even this case. Penzl (1975, §4.5a) writes: 'Die Forschung hat diesen Wandel von **e* zu **i* unter der Einwirkung von *i*-Lauten der Folgesilbe (Nasalen oder Akzentschwäche) als *i*-Umlaut bezeichnet.' For **e/* > **i/* in unaccented syllables, cf. fn. 40, below.

⁶ The phonemic stage **o/* is obviously assumed only for North and West Germanic (see, e.g., Penzl, §4.5b). Intermediate stages, e.g. **[ɛ]* and **[i]*, have been postulated also for **e/* > **i/* (cf. Penzl, §4.5a) and **i/* > **e/* (cf. Hock 1973, §10.1). But the view that these changes were

except when a high front semivowel or a cluster of nasal plus consonant intervenes (*a*-umlaut): PIE */wiros/ (cf. Lat. *vir*) > PGmc. */wiraz/ > OIc. *verr*. OE OFris. OS OHG *wer* (Go. *wair*) 'man';⁷ PIE */krnom/ (cf. Skt. *śṛṅgam*) > PGmc. */hurnan/ > OIc. OE OFris. OS OHG *horn* (Go. *haur̥n*) 'horn'—but PIE */kuntos/ (cf. Gk. *kun-*) > PGmc. */hundaz/ > OIc. *hundr*, OE OFris. OS *hund*, OHG *hunt* (Go. *hunds*) 'hound, dog'.

However, while the *a*-umlaut of */i/ to */e/ as a PGmc. phenomenon has always been contested,⁹ and another change—that of */e/ to */i/ before high back vowels (*u*-umlaut)—has also been ascribed to Proto-Germanic,¹⁰ the view has more recently been advanced that since both the 'earlier umlauts' and the 'later umlauts' are directly related to the Germanic accent shift, at least the phonetic stages of all these vowel modifications, produced by both vocalic and consonantal influences, should be ascribed to Proto-Germanic.¹¹

But unless one is prepared to multiply hypotheses to the highest degree, this attractive theory can hardly be reconciled with the available evidence. More specifically, even the testimony of the North and West Germanic languages makes it extremely difficult to assume a single allophonic stage traceable to Proto-Germanic, since neither the 'umlaut' results nor the basic tendencies are the same in all cases. This is clearly illustrated by the operation of *u*-umlaut in the individual dialects: (1) In Old Saxon and Old High German, */-u/ had a raising influence on */e/—cf. *filu* < */felu/ (OFris. *felo*) 'much'. (2) In Old Icelandic, it had a rounding influence on */i e a/ among the short vowels—cf. *miklu/myklo* 'much, by far',

completed through 'a reassignment of allophones, according to which [ɛ] was attracted into the orbit of [i] (e.g. */seti-/ = */[seti-] > */siti-/ and [i] into the orbit of [e] (e.g. */nista-/ = */[nista-] > */nesta-/)', as stated by Antonsen (1965:28; cf. 1964:183), is hardly tenable, since the further raising and lowering were undoubtedly brought about not so much by the relatively small space between [ɛ] and [i] and by the asymmetry of the short/lax subsystem (see Antonsen 1964:183) as by the influence of the following umlaut determinants */-i/ and */-a/. However, the assumption that */i/ developed an allophone */[i]/ only after the merger of */e/ with */i/ (see Hock, §10.1) is quite unnecessary, since the raised allophone of */e/ may well have been completely fronted (*/[e]/) and the lowered allophone of */i/ somewhat retracted (*/[i̠]/), with a sufficient 'margin of safety' for 'the alleged exchange' (Hock, §6.4) to have taken place.

⁷ Cf. Eng. *wer*[e]wolf (Ger. *Werwolf*). Other relevant examples are PIE */nisdos/ (cf. Lat. *nidus*) > PGmc. */nistaz/ > OE OS OHG *nest* 'nest'—and, perhaps, PIE */imnos/ (cf. Lat. *imāgo*) > PGmc. */ibnaz/ > OIc. *iafn* > *iamm* ('*a*-breaking' of */e/), OE *efen* > *emn*, OFris. *even*, OS *eban*, OHG *eban* (Go. *ibns*) 'even'. That */[i] had already acquired phonemic status in (late) Proto-Indo-European is indicated by the dative singular of the Gmc. *i*-stems, which should be derived from PIE */-eji/ and */-oji/ (cf. Antonsen 1972:138). But the phonemization of PIE */[i u] can hardly have occurred, as I will argue in a separate article, 'through "system balance" upon the rise of */i/ and */ū/' (Antonsen 1972:138).

⁸ For the use of [h] to designate a voiceless velar spirant, on the model of [g] for its voiced counterpart, see Cercignani (1979d, §1).

⁹ See, e.g., Paul (1879:82–4), Kock (1898:553–4), and Krahe–Meid (1969, §36). For Lloyd's rejection of the change as such, see §2 and fnn. 37, 39, below.

¹⁰ See Marchand (1957:350–53). For a rejection of the view that the 'earlier umlauts' turned the opposition */i/ ≠ */e/ into a variation */i ~ e/, see Cercignani 1979c.

¹¹ See Antonsen (1961:218–20, 1965:25–8, 1972:140). Note that Antonsen's 'umlaut' determinant */-u/ also covers a following velar consonant (1964:181, fn. 14, 1967a:8).

-[f]reþr/-[f]røþr 'peace',¹² and *mōgr* < */maguz/ (Go. *magus*) 'boy'.¹³ (3) In Old English, it had a diphthongizing influence on */i e æ/—cf. *silofr/siolufr* > *sio/seolf* 'silver', *medu/meodu* 'mead', and Mercian *hæfuc/heafuc* (OE *hafoc*) 'hawk'. (4) In Old Frisian, it had a diphthongizing influence on */i/—cf. *nigun/niugun* 'nine'. One might object that the OS and OHG forms are much older than those of, say, Old Icelandic, and that the traditional interpretation of the OE digraphs as diphthongs is by no means undisputed.¹⁴ But the fact remains that none of these languages favors the postulation of a backing effect of */-u/. The assumption that the *u*-umlaut of */e/ to */i/ in Old Saxon and Old High German resulted from the coalescence of a back allophone of PGmc. */e/ with a back allophone of PGmc. */i/, which subsequently merged with the primary (front) allophone of */i/ itself,¹⁵ has nothing to support it except the necessity of reconciling the claim that */-u/ invariably had a backing influence with the circumstance that no such influence is in fact attested in Old Saxon and Old High German. Similarly, the hypothesis that the Olc. rounding was only a secondary development¹⁶ is hardly supported by the available evidence.¹⁷ As for the view that the OE 'short' digraphs *io eo ea* denote monophthongal phonemes reflecting earlier back allophones,¹⁸ this again necessitates the postulation of secondary changes to obliterate the alleged backing effect, sometimes even when */-u/ is still present. For example, PGmc. [a] has been said to change to back [ɑ] only to revert to central [ə] in Pre-English (OE *ea*, as in *featu* 'vats'), on the assumption of a 'spontaneous' development of allophones which are supposed to have occurred only 'before back vowels and velar consonants'.¹⁹ Again, PGmc. [e] has been said to change to back [ɛ] (OE *eo*, as in *meodu* 'mead') only to revert to central [ɛ̃] = *eo ue oe o u* in early Middle English, and to front [e] = *e* in later Middle English. The assumption here is that the central nature of /ɛ̃/ is attested 'by its later development in Middle English:

¹² The /e/ of -[f]reþr is best explained by lowering of */i/ (*friþr* < */fripuz/) in the second element of compound names (cf. Hock, 331). But OFris. *fretho*, OS *freðu* vs. OE *fri[o]þu*-, OS *friðu*, OHG *fridu* reflect */i/ > */e/ before */-a/ in the genitive singular (*/fripawez/).

¹³ Even if 'w-umlaut' is left out of account, the rounding influence of */-u/ on */i e/ is attested at least after labials (see Heusler 1932, §§67–75 and, for a different presentation, Noreen 1923, §§77, 82). Cf. also the 'u-breaking' of */e/, as in *miþr* (OE *me[o]ðu*) 'mead', which is explained either as *e* > *ea* > *ja* (breaking, shift of stress) > *jə* (*u*-umlaut), or else as *e* > *eu* > *eo* > *jə* > *jə* (breaking; see, e.g., Gordon 1957:274).

¹⁴ For the relevant references, see the discussions in Ney 1969 and Giffhorn 1974 (both in favor of the traditional interpretation).

¹⁵ See Antonsen (1964:187), where OHG *frithu/fridu* 'peace' is erroneously given with original */e/ instead of */i/ (cf. Antonsen 1965:26, but 1961:230).

¹⁶ See Antonsen (1961:220–24), with a discussion of earlier literature on the subject (and cf. Antonsen 1963:203–4, 1967b: 25, fn. 25).

¹⁷ Cf. Benediktsson (1963:420). Haugen (1969:114–5) speaks of 'definitely gratuitous assumptions about Icelandic'.

¹⁸ See Antonsen (1961:224–9, 1967a). His interpretation of these digraphs is based on Hockett's (1959:575–6).

¹⁹ See Antonsen (1967a:9–10, and cf. 1961:225, 228–9); he further assumes central [ə] to have become eME [æ] = *æ* (> ME [a] = *a*). The whole series of changes proposed by Antonsen (PGmc. [a] > [ɑ] > PE [ə] > eME [æ] > ME [a]) is thus more complicated than would appear from Ney's discussion (45). For the development of ME /a/, cf. also Cercignani (1980, §§33, 39).

normally it coalesces with front spread /e/, but when found in environments conducive to rounding, it falls together with back round /o/, e.g. OE /wārpan/ > ME /worpən/ 'throw'.²⁰ But apart from the fact that the East Midland and Southern form of this verb has /u/, both *wurpen* and *worpe[n]* (which is chiefly Northern) reflect changes that originated in Old English—cf. IWS *wurpan* and INhb. *worpa* (OE *weorpan*).²¹

2. THE 'EARLIER UMLAUTS' AS IDIOLINGUAL PHENOMENA. While no 'umlaut' phenomena can be assumed for Proto-Germanic because of the idiosyncratic situation of Gothic,²² both the 'earlier' and 'later' vowel modifications in question should be ascribed to the prehistory of the individual languages, rather than to a 'Northwest Germanic' proto-language of the third order. For, in spite of remarkable similarities in both trends and results, even the allophones produced by the 'earlier' phenomena may well have developed at different times (and under different conditions) in the various North and West Germanic languages. Thus the change */e/ > */i/ in accented syllables occurred before *[-i] and before clusters of nasal plus consonant in all the relevant dialects; but in some of them, it also took place before *[-u]—cf. OS OHG *filu* < */felu/ (OFris. *felo*) 'much'—and before single nasals regardless of the following vocalism—cf. OE *niman*, OFris. *nima/nema*, OS *niman/neman*, OHG *neman*, Olc. *nema* (Go. *niman*) < */nemanan/ 'take'.²³ Of course, one may insist that a 'geographically limited' change like */e/ > */i/ before *[-u] was 'a late innovation of Continental West Germanic';²⁴ but the truth may well be that there was only a single raising process of */e/ to (*[ē] >) */i/, in partly different environments, in the prehistory of each dialect.²⁵ Similar considerations apply, mutatis mutandis, to the change */u/ > */o/ (> */o/) before *[-a], which was prevented by an intervening cluster of nasal plus consonant in all the North and West Germanic languages, but also by an intervening single nasal in some dialects—cf. OE *cuman*, OFris. *kuma/koma*, OS *kuman*, OHG *koman/kuman*, Olc. *koma*

²⁰ Antonsen (1961:228–9), where central /ê/ obviously replaces the traditional front /ö/.

²¹ For these forms, see Campbell (1959, §§147, 321); Brunner (1965, §113); and cf. Jordan–Matthes (1934, §66, fn. 3), to which Antonsen refers.

²² For a rejection of the highly speculative view that Gothic originally exhibited the same 'earlier umlauts' as the other Germanic languages, see Cercignani 1979a (and cf. fn. 1, above).

²³ The forms with /i/ could perhaps be explained as analogical, 'da alle anderen starken Verba mit *u* im Partizip im Präsensstamm *i* haben' (Luick 1914–40, §74, n. 2; cf. Helten 1909:124). But the influence of a single nasal is also seen at work in the case of */u/ (see below, and cf. Kufner 1972:89).

²⁴ Hock (329). The fact that OS OHG */e/ > */i/ also occurred before *[-u] representing PGmc. */-ō/—e.g. in OS *gibu*, OHG *gibu* < [gebō] (OE *geofu/giefe*) 'I give'—need not imply that 'the change is relatively late' (Campbell, §113, fn. 7; cf. Cercignani 1973:106) or that a chronological distinction should be drawn between an older *u*-umlaut of */e/ to */i/ and 'the change of *e* > *i* before secondary *u* (< *ō*)' (Benediktsson 1967:191, fn. 30). As to Runic *gibu* for expected *gebu*, it should be explained as reflecting */e/ > */i/ before *[-i] in the 2nd and 3rd persons singular (cf. Krause 1971, §36.2, and Antonsen 1975:22). The change */-ō/ > */-ū/, proposed by Antonsen 'finally and before a high vowel with or without an intervening */n/' (1972:139), cannot of course be regarded as late Proto-Germanic, but at most as 'Northwest Germanic' (cf. Antonsen 1969:58, 60, 1975:16).

²⁵ Cf. Luick (§74), where it is suggested that *e* > *i* before *m* might be contemporaneous with *e* > *i* before the other determinants.

(ONorw. *koma/kuma*) < */k^wumanan/ 'come'; OE *punor* (OFris. *thun/thonresdei* 'Thursday'), OS *thunar* (MLG *dunre/donre*), OHG *donar*, OIc. *pōrr* < */punaraz/ 'thunder'.²⁶ Nor is it unlikely that in some dialects the change in question was also prevented by a neighboring labial—cf. OE *full*, OFris. OS *ful/fol*, OHG *fol*, OIc. *fullr* (Go. *fulls*) < */fullaz/ 'full'; OE *fugol*, OFris. *fugel*, OS *fugal/fogal*, OHG *fogal/fugal*, OIc. *fugl/fogl* (Go. *fugls*) < */fuglaz/ 'fowl, bird'.²⁷ For while, in certain dialects and individual forms, /u/ for 'expected' /o/ may well result from paradigmatic leveling, analogical replacement alone cannot adequately account for the fact that /u/ for /o/, or the variation between /u/ and /o/, is especially frequent (at least in some dialects) in labial contexts.²⁸

Again, the change */i/ > */e/ before *[-a] appears to have been influenced by the consonantal environment;²⁹ and although the original conditions under which the lowered and retracted allophone *[i] failed to develop in some dialects are hard to recover, those which are usually indicated for North Germanic cannot be assumed for West Germanic. In Old Icelandic, PGmc. */i/ was apparently retained after */k/ and */g/;³⁰ but this was by no means true of at least certain types of Old High German, the language which shows the largest number of forms with /e/ from */i/³¹—cf. OIc. *skip* vs. OHG *skif/skef* (< */skipan/) 'ship'; OIc. *gin* (< */ginan/) 'mouth (of beast)'³² vs. OHG *ginēn/genēn* (*giwēn/gewōn*) 'yawn'. Analogical replacement must, of course, be responsible for various forms with /i/ for 'expected' /e/. Thus the /i/ in the past participle of the first strong class (cf. OHG *rītan* 'ride', *reit*, *ritum*, *giritan*) can be explained as reflecting the combined influence of the preterit plural and of the *i*-timbre of the present.³³ To be sure, the /i/ of OHG

²⁶ Cf. Luick (§81), where retention of *u* is preferred to a late reversion of *o* to *u* before single nasals.

²⁷ See, e.g., Bülbring (1902, §116, fn.); he suggested that *u* might have been retained in labial contexts. Bülbring's alternative explanation that *u* was first lowered and then raised again (cf. Wright & Wright 1925, §§108–9) should of course be rejected. But while the (dialectally analogical) /o/ of words like OE *folc*, OFris. *folk* (*fulk*), OS OHG OIc. *folk* (< */fulkan/) 'folk, people' does not speak against a retention of */u/, the /u/ of words like *full* can hardly be taken (despite Luick, §78, fn. 3) to reflect a sort of paradigmatic leveling assisted by the consonantal environment.

²⁸ See, e.g., Brunner (§58), and cf. Holthausen (1921, §88, fn. 1: 'Hier sind wohl teils dialektische Eigentümlichkeiten, teils Einflüsse benachbarter Laute, teils Ausgleichungen der Grund des Schwankens.')

²⁹ Cf. Coetsem (1956, §22)—where, however, the development in question is regarded as Proto-Germanic because of its 'general diffusion' (but cf. fn. 22, above).

³⁰ See Noreen (§60) and Gutenbrunner (1951, §26.I.2), to which Coetsem (1956, §22) refers.

³¹ For a list of such forms, see Schatz (1927, §9) and cf. Braune-Mitzka (1967, §31, fnn. 1–3).

³² The example with */g/ given in Gutenbrunner (§26.I.2) is *gil* 'ravine'. But this is a neuter *ja*-stem (*/giljan/) and is therefore irrelevant.

³³ Cf. Prokosch (1939, §41a) and Hock (340–41). No such influence need be assumed, however, for the past participle of OIc. *vega* 'fight', an aorist-present verb with */i/ instead of */i/ (Go. *weihan*, OHG *wihan/wigan*, OE *wigan*), which was transferred from the first to the fifth class because of formal identity not only of its infinitive and past participle, but also of its preterit singular (*vā* < */wa'h-/), with the corresponding forms of *vega* 'weigh': *vā* (< */wa'h/ < */waq-/), *vōgom*, *vegenn* (cf. Prokosch, §54c). The ONorw. variant *viga* 'fight' can have /i/ from retained */i/ before *[-i] in the 2nd and 3rd persons present singular.

nidar[e] (< */niþera-/) beside Olc. *neþre* (*niþre*), OE *neopor* (*niþor*) 'nether, down' might perhaps derive from a form */niþiri/ < */niþeri/,³⁴ possibly with some influence on OHG *nidana* beside Olc. *neþan*, OE [be]neopan (*niþan*) 'beneath'. But such instances as OHG *hina*, *hinana* beside Olc. *heþan*, OE *heonan* (*hionan*, *hin-*) 'hence' and OE *hider* (< */hiðran/) beside Olc. *heþra* (*hipra* < */hiðrōn/) 'hither' can hardly be accounted for satisfactorily simply by recourse to the analogy of OHG *nidar*[e] etc.—or, even worse, to that of OHG *ubar* beside *obar* 'over'.³⁵ Nor is it likely that Olc. *fiskr*, OE *fisc*, OFris. OS OHG *fisk* (Go. *fisks*) < */fiskaz/ 'fish' owes its /i/ to the fact that an original */fiskiz/ 'was remade into an *a*-stem only after the phonemicization of *a*-umlauted vowels had been completed'.³⁶ As in other cases (e.g. OHG *skif* beside *skef*), the explanation may well be partly analogical. But the assumption of 'a type of systemic analogy', instead of an *a*-umlaut of */i/,³⁷ is by no means convincing, since it cannot be reconciled with the fact that an *a*-umlaut of */u/ did take place; and the theory would at any rate have to be re-interpreted as presupposing an analogical substitution of the alternation */e/ ~ */i/ for the non-alternation */i/ ~ */i/ in the prehistory of the individual dialects—cf. OHG *fel* (< */fellan/) 'skin' ~ *fillen* (< */fill-/ < */felliþanan/) 'to skin'; OHG *nest* (< */nistaz/) 'nest' ~ *nisten* (< */nistijanan/)³⁸ 'to nest, live' for an 'expected' OHG *nist* ~ *nisten*.³⁹ The type of analogy which must have applied here is paradigmatic leveling, most likely from a genitive singular in */-es/, before which the original vowel was retained.⁴⁰ But the extreme scarcity of forms with /e/ by *a*-umlaut of */i/ seems to imply that the assimilation exerted by */-a/ on */i/—being less powerful than that exerted by */-i/ on */e/—was resisted, with varying results, in order to avoid a merger of */i/ with */e/: in some types of North and West Germanic, */i/ = [i] was kept apart from */e/, in others it was identified with the lower phoneme. In the case of */u/ before */-a/, however, the early allophone */u/ invariably became */o/, since there was no */o/ with which to avoid confusion. And when the umlaut determinant */-a/ was lost in certain positions,

³⁴ Cf. OHG *after*, OE *æfter* beside Olc. *eptir* 'after'—which Antonsen (1975:44) derives, respectively, from */aftera-/ and */afteri/.

³⁵ See Hock (342), where *obar* is said to show 'generalization of the alternant found in *obana*'. Note that OHG *ubar* (> MHG *uber*) beside OHG *obar*, OS *obar*, OFris. *over*, OE *ofer* 'over' should be derived from */uber/ beside */uberi/ > */ubiri/ > Olc. *yfir*, OHG *ubir*[i] > MHG *uber*; and that OE *ufan* beside Olc. *ofan*, OFris. *ova*, OS *oban*[a], OHG *obana* (< */uban-/) is patterned on [on] *bufan* (< */ubanan/ < */be-uban-) 'above', which has a labial both before and after the /u/.

³⁶ Hock (343), where Lat. *piscis* is adduced as supporting evidence.

³⁷ See Lloyd 1966. For an earlier rejection of this change, see Bremer 1909.

³⁸ In Notker, beside *nestōn* (< */nistōnan/).

³⁹ Lloyd (738) claims that 'the partial overlapping in Germanic of the two phonemes /i/ (represented in all environments by [i]) and /e/ (with the allophones [i] and [e]) led to the occasional development of an *e*-allophone of /i/ by a type of systemic analogy'.

⁴⁰ Cf. Antonsen (1972:123). The available material is not entirely conclusive; but in the absence of evidence to the contrary, it would seem reasonable to assume that the change of unaccented */e/ to */i/ occurred in all positions in Pre-Gothic, and only in favorable contexts (including before final */z/) in the prehistory of the individual North and West Germanic languages (cf. Cercignani 1979c).

*[o] was phonemized to */o/, while *[i] was merely subsumed (insofar as it had escaped further lowering) under the higher allophone of */i/.

Thus neither the 'earlier umlauts' nor the 'later umlauts' should be ascribed to Proto-Germanic—or, for that matter, to 'Northwest Germanic'. True, the disruption of the autonomy of the IE syllable, as a result of the Germanic shift and fixation of the accent on root syllables, was undoubtedly a prerequisite for all umlaut phenomena; but this does not necessarily imply that it was also their immediate cause.⁴¹ Indeed, the evidence of those dialects in which umlaut phenomena can actually be shown to have taken place indicates that even their earliest phonetic stages should be ascribed to the prehistory of the individual (North and West) Germanic languages, and that the similarities in certain trends and results should be explained from the development of similar prosodic features in languages exhibiting similar paradigmatic and syntagmatic structures. The gradual increase in strength of the Germanic stress accent, as attested by the progressive weakening and loss of vowels in unaccented syllables, was no doubt a decisive factor in this process. However, its effect was not so much a 'coloration transfer' from weakening unaccented vowels to accented ones,⁴² but rather an assimilation of accented vowels to the vowel or semivowel of a following syllable before the latter's reduction—or, more precisely, an anticipation, in the articulation of accented vowels, of at least one feature of the vowel or semivowel of the following syllable. Where such a gradual increase was hindered, or its prosodic effects counter-acted, by exceptional external factors, the result was (apart from consonantal influence) a more restricted type of vowel assimilation (e.g. only */e/ > */i/ before *[-i]),⁴³ or no vowel assimilation at all. This explains why the language of the Goths, who left Scandinavia for the shores of the Baltic and then migrated quite early to the Black Sea steppes, exhibits such an idiosyncratic situation with regard to umlaut phenomena.

3. A PARALLEL CASE: NASALIZATION AND COMPENSATORY LENGTHENING. That the 'earlier' vowel modifications produced by consonantal influences should likewise be ascribed to the prehistory of the individual Germanic languages, and their similarities explained from the development of similar prosodic features in related dialects, is shown not only by the changes that have just been discussed, but also by those connected with the nasalization and compensatory lengthening of */i a u/. As is well known, Proto-Germanic is usually assumed to have had the long nasalized vowels *[ī ā ū] before */h/,⁴⁴ on the ground that all the Germanic languages show loss of */n/ before */h/ with compensatory lengthening (and nasalization) of the preceding vowel—cf. Go. *peiħan*, OE *þion/þēon* ~ *ġepungen*, OS *thīhan* ~ *githungan*, OHG *dīhan* (< */pinħanan/) 'thrive ~ excellent';⁴⁵ Go. *fāhan*, OIc. *fā* ~

⁴¹ Haugen (1969:107) writes: '[umlaut] is often attributed to the fixing of stress on the tonic syllable in Proto-Germanic, but it is dubious usage to speak of it as a CAUSE'.

⁴² See Höfler (1955:62), where the process is described as 'eine Umfärbung in die Richtung auf die Qualität, die "Farbe" der farbverlierenden Nebentonvokale'.

⁴³ For a hypothetical postulation of this change in Pre-Gothic, see Cercignani 1979a.

⁴⁴ See, e.g., Streitberg (§93), where one finds the often repeated observation that, since *[ā] appears as *ō* in Old English, the nasalization itself must have been lost in the individual languages.

⁴⁵ Also OIc. *þēttir* (< */þiħtt-), OFris. **thlicht* 'tight, thick' (cf. WFrís. *ticht*, with shortening before the consonant cluster).

fangenn, OE *fōn* ~ *fangen*, OFris. *fā*⁴⁶ ~ *fangen*, OS OHG *fāhan* (< */fanhanan/) ~ *gifangan* 'seize ~ seized'; and Go. *pūhta* ~ *þugkeiþ*, OIc. *þōtta* (< */pūht-/) ~ *þykke* (< */þunk-/), OE *þūhte* ~ *þyncþ*, OFris. *thūchte* ~ *thenszt*, OS *thūhta* ~ *thunkid*, OHG *dūhta* (< */punhtōn/) ~ *dunkit* '[me] thought ~ [me] thinks'. But the loss of */n/ before */h/ can hardly be dissociated from a similar loss of both */n/ and */m/ before other voiceless spirants (*/f s þ/) in the so-called Ingvaconic languages (including the partly 'Franconized' OS variety) and, with certain restrictions, in Old Icelandic—cf. OE OFris. OS *fif* vs. OHG *fimf*/*finf*, Go. *fimf* (< */femfe/) 'five'⁴⁷; OE OFris. *gōs*, MLG *gās/gōs*, OIc. *gōs* vs. OHG *gans*, Go. **gansus*⁴⁸ (< */gansuz/) 'goose'; and OE *mūþ*, OFris. *mūth*, OS *mūd* vs. OS OHG *mund*, Go. *munþs* (< */munþaz/) 'mouth'.⁴⁹

To be sure, the loss of */n/ before */h/ and the loss of */n m/ before other voiceless spirants need not reflect a single phenomenon in all the relevant dialects, as indeed certain 'Ingvaconisms' in the modern Alemannic dialect (*fif*, *gās/gōs* etc.) would seem to indicate.⁵⁰ But as this may well have been the case in, say, Pre-English or Pre-Frisian, it would obviously be unsafe to assume that the changes connected with the loss of */n/ before */h/ were necessarily earlier than those connected with the loss of */n m/ before other voiceless spirants. The nasalization and compensatory lengthening of */i a u/ should therefore be ascribed to the prehistory of the individual Germanic languages, and the similarities in their results explained from the development of similar prosodic features in dialects exhibiting similar paradigmatic and syntagmatic structures. Thus neither the allophones */ĩ ā ũ/⁵¹ nor the phonemes */ĩ ā ũ/⁵² should be postulated for Proto-Germanic. All that we need to do to account satisfactorily for the attested phenomena is to assume a slight ('normal') nasalization for all PGmc. vowels in a nasal environment, and an idiolingual, non-uniform increase in the degree of nasalization according to language as well as position. Where the increase occurred only before [ŋ]—whose oral resonance is much more restricted than that of [n m]⁵³—the result

⁴⁶ From */fwā/ < */fūa/ < */fōa/ < */fōhan/ (see Steller 1928, §12, and Krogmann 1970:204).

⁴⁷ Also OE *fifel*, OIc. *fifl* < */fimfil-/ < */femfil-/ 'monster' beside OIc. *fimbul-* 'huge'. OIc. *fim[m]* for **fif* 'five' is patterned on the ordinal *fimte*, with loss of the interconsonantal */f/ (Go. *fimfta*; see Noreen (§§291.4, 298.2).

⁴⁸ Implied by Sp. *ganso*.

⁴⁹ OIc. *mubr* is from *munnr* < */munþ-/; and OIc. *mēl*, OSwed. *mīl* (OE *mīþl* but OHG *mindil*) is from */minnl-/ < */minþl-/ < */menþil-/ (see Noreen, §§110.1, 261, 275). Yet the latter is often given as a relevant example (cf. Kufner, 89, Moulton 1972:170).

⁵⁰ See, e.g., Kufner (93), where the prehistoric 'Ingvaconic' change is said to have taken place 'centuries later in much of Alemannic'.

⁵¹ See Coetsem (1956, §50, 1970:54), Moulton (1961:13; no opposition between, say, */āh/ and */ānh/); and Antonsen (1972:127). Penzl (§§4.7, 4.8a) writes that PGmc. */ā ī ũ/ before */h/ 'were still phonemically */ān in ũn/' (by which he seems to mean */an in un/), and that */ā < */ā̃/ became an allophone of */ē/ 'auf die Stellung vor geschwächtem velarem Nasal beschränkt'. Reis (1974, §§2.1.2.2, 3) would analyse */ā̃/ (by which she seems to mean */ā̃/) as */an/, but would assign */ĩ ũ/ to */i ũ/—with a phoneme */ā̃/ occurring only in unaccented syllables.

⁵² See Moulton (1972:170), where the loss of */n/ is regarded as sufficient to produce three new phonemes which occurred only before */h/.

⁵³ Cf. the not infrequent use of (relatively) oral vowels plus /ŋ/ in, e.g., Ger. *Restaurant* (Fr. /ã/), *Refrain* (Fr. /ẽ/), *Parfum* (Fr. /œ/) etc.

was a loss of */n/ before */h/; where it occurred before [ŋ n m], the result was a loss of */n m/ before any voiceless spirant, provided that the loss itself was not prevented by special assimilatory changes, such as */nþ/ > */nn/ in Old Icelandic—cf. Olc. *annarr* (OS OHG *andar*, Go. *anþar* < */anþeraz/) vs. OE *ōper*, OFris. *ōther*, OS *āðar/ōðar* ‘other’. That nasalization became particularly strong in the Ingvaëonic languages and in Old Icelandic is of course confirmed by the OE and OFris. treatment of PGmc. */a ē/ before nasals—cf. OE OFris. *man[n]/mon[n]* ‘man’ and *mōna* (Go. *mēna*) ‘moon’—as well as by the evidence provided by the anonymous Icelandic author of the 12th century *First grammatical treatise*.⁵⁴

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⁵⁴ See Haugen (1972:38–9) and Benediktsson (1972, §5.2.2.2a), where further evidence is adduced in favor of a Nordic distinction between nasal and oral vowels.

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